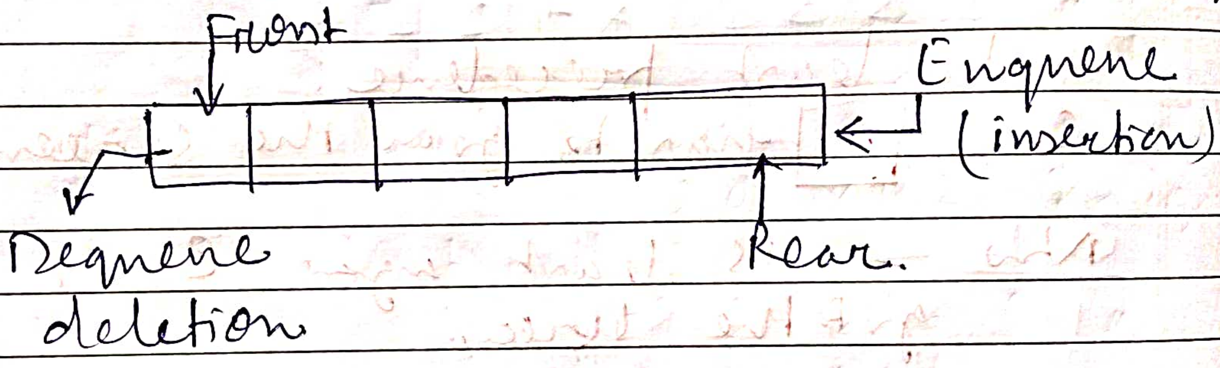
Characteristics of Queue

- Handle multiple data.
- Both ends are accessible.

Queue - The name of the array storing elements.

Front - the index where the first element is stored in the array.

Rear - the index where the last element is stored in an array



Operations of Queue

Enqueue
Dequeue
IsEmpty - Returns True if queue is empty.
IsFull - Returns true if the queue is full.

Peek - Get the value of the front of the queue without removing it.

Front is const. it doesn't move, for any Enqueue or Dequeue operation Rear gets updated or decremented respectively

ALGO

Initial value of FRONT = REAR = -1.

For 1st element to insert, Front = 0 = Rear

For Enqueue

Enqueue(2)

Enqueue(10) F = 0, R = 0+1 = 1

At 1st posn insert 10

0	1
2	10

- Check if Queue is full. ($\boxed{Rear = Max - 1}$) overflow.
- For first element, set FRONT = 0.
- Increase Rear by 1
- Add new element in the position pointed by Rear

Front → 0, 1 ← Rear

2	3	

Enqueue(5) → update Rear++

New Rear = 1+1 = 2

So at 2nd posn we can ~~add~~ insert 5.

Dequeue

- Check if Queue is Empty. ($\boxed{F = R = -1}$)
- Return the value pointed by FRONT.
- increase FRONT by 1.
- Return success.

0	1	2	3	4
2	10	-1		6

F, R

Dequeue( ) = Delete 2 & front++ ~~Dequeue++~~

So now front = 0+1 = 1

Drawback of Linear queue.

To insert we need to check for overflow condition. i.e ~~front~~ as Rear = Max − 1

0	1	2	3	4	
2	3	5	6	7	~~8~~

f → (index 0), R → (index 4)

Suppose size of Array = 5.
So Rear = 5 − 1 = 4
front = 0.

Now you call for enqueue(−1) = Not possible at R = 4 the max limit.

Suppose I performed Dequeue( ).

Now

0	1	2	3	4
	3	5	6	7

f → (index 1), R → (index 4)

F = F++ (after deletion)
= 1 & R = 4.

Still here enqueue(−1) not possible cause the Rear where insertion has to be made is full.

To overcome this problem we got Circular queue where Rear points back to front.

0	1	2	3	4
	3	5	6	7

↑F (index 1), ↑R (index 4)

Now F = 1 & R = 0, enqueue(−1) possible.
F = 1 and R = 0

Here overflow condⁿ is checked as

F = 1, R = 0 Full. ((rear + 1) % N) = = Front

F = 1 ((R + 1) % 5) = = Front